Linear Algebra with Applications in Automatic Control

School of Automation,
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Fall Semester

Outline

- Chapter 2- Polynomials and matrix polynomials
  - Part 1. Polynomials –the first four sections
  - Part 2. Matrix polynomials –the last four sections
- Today we go into details of Section 2.1 -Section 2.4
  - S1. Linear algebras
  - S2. Polynomial ring and Euclidean division
  - S3. Ideals of polynomials
  - S4. Factorization of polynomials

2.1 Linear algebras

Defintion 2.1 A linear space $\mathcal{V}$ over a field $\mathcal{F}$ is referred to as a linear algebra if the binary operation $\circ$, called multiplication, is closed on $\mathcal{V}$, i.e. for any $(\boldsymbol{\alpha}, \boldsymbol{\beta}) \in \mathcal{V} \times \mathcal{V}$, there is a unique $\boldsymbol{\gamma} \in \mathcal{V}$ such that $\boldsymbol{\gamma} = \boldsymbol{\alpha} \circ \boldsymbol{\beta}$, and satisfies the following properties

1. multiplication is left and right distributive over addition, that is, for all α, β, $\gamma \in \mathcal{V}$ there hold

$$\begin{aligned} \boldsymbol{\alpha} \circ (\boldsymbol{\beta} + \boldsymbol{\gamma}) &= \boldsymbol{\alpha} \circ \boldsymbol{\beta} + \boldsymbol{\alpha} \circ \boldsymbol{\gamma}, \\ (\boldsymbol{\beta} + \boldsymbol{\gamma}) \circ \boldsymbol{\alpha} &= \boldsymbol{\beta} \circ \boldsymbol{\alpha} + \boldsymbol{\gamma} \circ \boldsymbol{\alpha}; \end{aligned}$$

2. multiplication $\circ$ and the scalar multiplication satisfy

$$(a\boldsymbol{\alpha}) \circ \boldsymbol{\beta} = \boldsymbol{\alpha} \circ (a\boldsymbol{\beta}) = a(\boldsymbol{\alpha} \circ \boldsymbol{\beta}), \quad \forall\, \boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathcal{V}, \quad a \in \mathcal{F};$$

3. multiplication $\circ$ is associative, that is, for all α, β, $\gamma \in \mathcal{V}$ there hold

$$\boldsymbol{\alpha} \circ (\boldsymbol{\beta} \circ \boldsymbol{\gamma}) = (\boldsymbol{\alpha} \circ \boldsymbol{\beta}) \circ \boldsymbol{\gamma}.$$

2.1 Linear algebras

If in addition, there is an $\epsilon \in \mathcal{V}$ such that

$$\epsilon \circ \alpha = \alpha \circ \epsilon = \alpha\,, \quad \forall \alpha \in \mathcal{V}\,,$$

then ϵ is said to be the unit in $\mathcal{V}$, and $\mathcal{V}$ is said to be a linear algebra with unit ϵ over field $\mathcal{F}$. If the multiplication is commutative in $\mathcal{V}$, that is,

$$\alpha \circ \beta = \beta \circ \alpha\,, \quad \forall \alpha\,,\ \beta \in \mathcal{V}$$

then $\mathcal{V}$ is called a commutative algebra.

2.1 Linear algebras

- Definition of isomorphism algebras

Definition 2.2 Let $\mathcal{U}$ and $\mathcal{V}$ be two linear algebras over $\mathcal{F}$. If

1. there is an isomorphic transformation $\mathscr{I} : \mathcal{U} \mapsto \mathcal{V}$,

2. with respect to this isomorphism there holds

$$\mathscr{I}(\boldsymbol{\alpha} \circ \boldsymbol{\beta}) = \mathscr{I}(\boldsymbol{\alpha}) \circ \mathscr{I}(\boldsymbol{\beta}), \quad \forall \boldsymbol{\alpha}, \ \boldsymbol{\beta} \in \mathcal{U}$$

then we say that $\mathcal{U}$ and $\mathcal{V}$ are isomorphic algebras.

2.1 Linear algebras

- Properties of linear algebras

Theorem 2.1 Let $\mathcal{V}$ be a linear algebra over the field $\mathcal{F}$ with the unit $\boldsymbol{\epsilon}$, and $a(\lambda) \in \mathcal{F}[\lambda]$. It is prescribed that for any $\boldsymbol{\alpha} \in \mathcal{V}$, $\boldsymbol{\alpha}^0 = \boldsymbol{\epsilon}$. Then for any $\boldsymbol{\alpha} \in \mathcal{V}$, $a(\boldsymbol{\alpha})$ corresponds uniquely to the element $a(\boldsymbol{\alpha}) \in \mathcal{V}$,

$$a(\boldsymbol{\alpha}) = \sum_{i=0}^{n} a_i \boldsymbol{\alpha}^i = a_n \boldsymbol{\alpha}^n + a_{n-1} \boldsymbol{\alpha}^{n-1} + \cdots + a_1 \boldsymbol{\alpha} + a_0 \boldsymbol{\epsilon}\,.$$

Theorem 2.2 Let $\mathcal{F}$ be a field, $a(\lambda)$, $b(\lambda) \in \mathcal{F}[\lambda]$. Then for any $\alpha \in \mathcal{F}$, both

$$\begin{aligned} x(\alpha) &\triangleq a(\alpha) + b(\alpha) \quad \text{and} \\ y(\alpha) &\triangleq a(\alpha) b(\alpha) \end{aligned}$$

are in $\mathcal{F}$.

2.1 Linear algebras

- Properties of linear algebras

Theorem 2.3 Let $\mathcal{V}$ be a linear algebra over $\mathcal{F}$ with unit, $a(\lambda)$, $b(\lambda) \in \mathcal{F}[\lambda]$, $\boldsymbol{\alpha} \in \mathcal{V}$, $\alpha_j \in \mathcal{F}$, $j = 1, 2$, then

1. $c(\lambda) = a(\lambda)b(\lambda)$ implies $c(\boldsymbol{\alpha}) = a(\boldsymbol{\alpha})b(\boldsymbol{\alpha})$, and

2. $c(\lambda) = \alpha_1 a(\lambda) + \alpha_2 b(\lambda)$ implies $c(\boldsymbol{\alpha}) = \alpha_1 a(\boldsymbol{\alpha}) + \alpha_2 b(\boldsymbol{\alpha})$.

2.2 Ring and Euclidean division

- Properties of Polynomial ring

Theorem 2.4 $a(\lambda),\ b(\lambda) \in \mathcal{F}[\lambda]$. If $a(\lambda) \neq 0$, $b(\lambda) \neq 0$, then

1. $a(\lambda)b(\lambda) \neq 0$;
2. $\deg a(\lambda)b(\lambda) = \deg a(\lambda) + \deg b(\lambda)$;
3. if both $a(\lambda)$ and $b(\lambda)$ are monic, then $a(\lambda)b(\lambda)$ is monic;
4. $\deg a(\lambda)b(\lambda) = 0$ if and only if $\deg a(\lambda) = \deg b(\lambda) = 0$;
5. if $a(\lambda) + b(\lambda) \neq 0$, then

$$\deg[a(\lambda) + b(\lambda)] \leq \max\{\deg a(\lambda), \deg b(\lambda)\}$$

2.2 Ring and Euclidean division

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- Properties of Polynomial ring-continued

Corollary 2.1 $\mathcal{F}[\lambda]$ is an integral domain. That is, for $a(\lambda)$, $b(\lambda)$, $c(\lambda) \in \mathcal{F}[\lambda]$ with $a(\lambda) \neq 0$, $a(\lambda)b(\lambda) = a(\lambda)c(\lambda)$, if and only if $b(\lambda) = c(\lambda)$.

- For integers, if a is divided by b, the we have quotient and remainder, i.e

Consider the set $\mathcal{Z}$ of all integers. If α, $\beta \in \mathcal{Z}$ and $\beta \neq 0$, then there are κ, $\rho \in \mathcal{Z}$ such that

$$\alpha = \beta\kappa + \rho$$

and either $\rho = 0$, or $0 < |\rho| < |\beta|$. κ and ρ are commonly called the quotient and the remainder of α divided by β respectively. Furthermore, for any $\beta \neq 0$, we have $|\alpha| \leq |\alpha\beta|$.

2.2 Ring and Euclidean division

Z. Song

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- Next, we shall extend these concepts to integral domain and introduce the concept of Euclidean domain and the associated quotient and remainder

2.2 Ring and Euclidean division

- Definition of Euclidean domain and the associated quotient and remainder concept

An integral domain $\mathcal{V}$ is referred to as a Euclidean domain if there exists a function $\nu : \mathcal{V}^* \mapsto \mathcal{Z}_{0,+}$ such that

1. for all $\boldsymbol{a}, \boldsymbol{b} \in \mathcal{V}$ with $\boldsymbol{b} \neq \boldsymbol{0}$ there are $\boldsymbol{q}, \boldsymbol{r} \in \mathcal{V}$ such that

$$\boldsymbol{a} = \boldsymbol{b} \circ \boldsymbol{q} + \boldsymbol{r} \tag{1}$$

and either $\boldsymbol{r} = \boldsymbol{0}$, or $\nu(\boldsymbol{r}) < \nu(\boldsymbol{b})$;

2. for all $\boldsymbol{a}, \boldsymbol{b} \in \mathcal{V}^*$ there holds

$$\nu(\boldsymbol{a}) \leq \nu(\boldsymbol{a} \circ \boldsymbol{b})\,. \tag{2}$$

where $\mathcal{V}^*$ the set of all nonzero elements in a ring $\mathcal{V}$.

Equation (1) is called a Euclidean division, $\boldsymbol{q}$ is the quotient and $\boldsymbol{r}$ is the remainder. $\nu(\cdot)$ is said a Euclidean function.

2.2 Ring and Euclidean division

- Next we shall extend Z to polynomials and show that polynomial ring is a Euclidean domain

Theorem 2.5 Let $a(\lambda), b(\lambda) \in \mathcal{F}[\lambda]$ and $b(\lambda) \neq 0$, then there exist unique $q(\lambda), r(\lambda) \in \mathcal{F}[\lambda]$ such that

$$a(\lambda) = b(\lambda)q(\lambda) + r(\lambda) \qquad (1)$$

and

$$\text{either } r(\lambda) = 0\,, \quad \text{or } \deg r(\lambda) < \deg b(\lambda)\,. \quad (2)$$

- Now, we define

$$\nu : \quad \mathcal{F}^*[\lambda] \mapsto \mathcal{Z}_{0,+}\,, \quad \nu[a(\lambda)] = \deg a(\lambda)$$

Corollary 2.2 $\mathcal{F}[\lambda]$ is a Euclidean domain.

2.2 Ring and Euclidean division

- Definition of quotient and remainder of polynomial ring

Theorem 2.5 Let $a(\lambda), b(\lambda) \in \mathcal{F}[\lambda]$ and $b(\lambda) \neq 0$, then there exist unique $q(\lambda)$, $r(\lambda) \in \mathcal{F}[\lambda]$ such that

$$a(\lambda) = b(\lambda)q(\lambda) + r(\lambda) \qquad (1)$$

and

$$\text{either } r(\lambda) = 0\,, \quad \text{or } \deg r(\lambda) < \deg b(\lambda)\,. \quad (2)$$

Definition 2.4 Equation (1) with $r(\lambda)$ satisfying (2) is called the normal of $a(\lambda)$ and $b(\lambda)$, $q(\lambda)$ and $r(\lambda)$ are called the quotient and the remainder of $a(\lambda)$ divided by $b(\lambda)$ respectively. Furthermore, if in (1), $r(\lambda) = 0$, then we say that $a(\lambda)$ is divisible by $b(\lambda)$, or $b(\lambda)$ is a divisor of $a(\lambda)$, denoted $b(\lambda)|a(\lambda)$.

2.2 Ring and Euclidean division

- Special properties of polynomial Euclidean domain

The uniqueness of $q(\lambda)$ and $r(\lambda)$ in the polynomial division is a special characteristics of $\mathcal{F}[\lambda]$. This is more than just the definition of a Euclidean domain. The uniqueness must not hold even for the motivating example $\mathcal{Z}$. κ and ρ will be unique if the division is restricted in the subset $\mathcal{Z}_{0,+}$ of $\mathcal{Z}$. However $\mathcal{Z}_{0,+}$ is not a ring.

2.3 Ideals of polynomials

- Definition of polynomial ideals

Defintion 2.8 Let $\mathcal{F}$ be a field, a subspace $\mathcal{I}$ of $\mathcal{F}[\lambda]$ is said to be an ideal if $a(\lambda)b(\lambda) \in \mathcal{I}$, $\forall a(\lambda) \in \mathcal{F}[\lambda]$, $b(\lambda) \in \mathcal{I}$.

From the definition it should be clear that a subspace $\mathcal{I}$ of $\mathcal{F}[\lambda]$ is an ideal if and only any multiplier of its any element belongs to $\mathcal{I}$.

- Definition of principle ideals and the generating element

Let $d(\lambda) \in \mathcal{C}[\lambda]$. Then

$$\mathcal{I}_d = \{\, x(\lambda) \;:\; x(\lambda) = d(\lambda)c(\lambda)\,, \quad c(\lambda) \in \mathcal{C}[\lambda] \,\}$$

is an ideal of $\mathcal{C}[\lambda]$. $\mathcal{I}_d$ defined in this way is called the principal ideal generated by $d(\lambda)$, $d(\lambda)$ is said to be the generating element of $\mathcal{I}_d$.

2.3 Ideals of polynomials

- Zero ideal and unit ideal

$\mathcal{I} = \{0\} \in \mathcal{F}[\lambda]$ Zero ideal.

$\mathcal{F}[\lambda]$ Unit ideal of $\mathcal{F}[\lambda]$.

- Unit ideal~~~~~ Generating element: a nonzero constant.

- Properties of polynomial ideal

Theorem 2.6 Let $\mathcal{I}$, $\mathcal{J} \subset \mathcal{F}[\lambda]$ be two ideals. Then the sum, denoted $\mathcal{I} + \mathcal{J}$, and the intersect, denoted $\mathcal{I} \cap \mathcal{J}$, of $\mathcal{I}$ and $\mathcal{J}$ are also ideals of $\mathcal{F}[\lambda]$.

2.3 Ideals of polynomials

- Properties of polynomial ideals-continued

Theorem 2.7 Let $\mathcal{I} \subset \mathcal{F}[\lambda]$ be a nonzero ideal, $d(\lambda)$ be the polynomial in $\mathcal{I}$ with the minimum degree, i.e. $\deg d(\lambda) = \min_{a(\lambda)\in\mathcal{I}} \{ \deg a(\lambda) \}$. Then $\mathcal{I}$ is the principal ideal generated by $d(\lambda)$. Furthermore, the monic generating element is unique.

Corrollary 2.3 $\mathcal{I} \subset \mathcal{F}[\lambda]$ is an ideal, then it must be a principal ideal.

- Key point: Euclidean integral domain

2.3 Ideals of polynomials

Z. Song

- Generalization of Theorem 2.7 to the context of Euclidean domain

Theorem 2.8 Let $\mathcal{V}$ be a linear algebra over the field $\mathcal{F}$. If there exists a Euclidean function $\nu : \mathcal{V} \mapsto \mathcal{Z}_{0,+}$ such that $\mathcal{V}$ becomes a Euclidean domain, then any ideal of $\mathcal{V}$ must be a principal ideal.

- Example 2.16 The set of stable transfer functions is a Euclidean domain, please refer to Pg 58.

2.3 Ideals of polynomials

- Definition of: Sum of ideals

Definition 2.6 Let $a_1(\lambda), a_2(\lambda), \ldots, a_r(\lambda) \in \mathcal{F}[\lambda]$, $\mathcal{I}_i$ be the ideal generated by $a_i(\lambda)$, $i = 1, 2, \ldots, r$. Then

$$\mathcal{I} = \mathcal{I}_1 + \mathcal{I}_2 + \cdots + \mathcal{I}_r$$

is referred to as the ideal generated by $a_1(\lambda)$, $a_2(\lambda), \ldots, a_r(\lambda)$.

From the definition we see that the ideal generated by $a_1(\lambda), a_2(\lambda), \ldots, a_r(\lambda)$ is the sum of all possible multipliers of the polynomials $a_1(\lambda), a_2(\lambda), \ldots, a_r(\lambda)$.

- Example 2.14 , Example 2.15 on Pg 56-57.

2.3 Ideals of polynomials

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- Relationship between generating elements of ideals and sum ideals

Theorem 2.9 Let $d_i(\lambda) \in \mathcal{F}[\lambda]$, $i = 1, 2, \ldots, r$, $\mathcal{I}_r$ be the ideal generated by $d_1(\lambda)$, $d_2(\lambda)$, ..., $d_r(\lambda)$ with the generating element $d(\lambda)$. Then

1. $d(\lambda)|d_i(\lambda)\ \forall i = 1, 2, \ldots, r$;

2. any $\gamma(\lambda)$ satisfying $\gamma(\lambda)|d_i(\lambda)\ \forall i = 1, 2, \ldots, r$ must satisfy $\gamma(\lambda)|d(\lambda)$.

2.3 Ideals of polynomials

- Intersect of ideals and the properties of the related generating element

Theorem 2.10 Let $a_i(\lambda) \in \mathcal{F}[\lambda]$, $i = 1, 2, \ldots, r$, $\mathcal{I}_i \subset \mathcal{F}[\lambda]$ be the ideal generated by $a_i(\lambda)$, $\mathcal{I} = \mathcal{I}_1 \cap \mathcal{I}_2 \cdots \cap \mathcal{I}_r$ with the generating element $a(\lambda)$. Then

1. $a_i(\lambda) | a(\lambda)$ $i = 1, 2, \ldots, r$;

2. any $b(\lambda)$ satisfying $a_i(\lambda) | b(\lambda)$ $i = 1, 2, \ldots, r$ must satisfy $a(\lambda) | b(\lambda)$.

Theorem 2.10 states that $\mathcal{I}_1 \cap \mathcal{I}_2 \cdots \cap \mathcal{I}_r$ is the totality of all possible common multipliers of $a_i(\lambda)$, $i = 1, 2, \ldots, r$.

2.3 Ideals of polynomials

Z. Song

- In the view of Theorem 2.9 and 2.10,
- we introduce

Defnition 2.7 Let $a_i(\lambda) \in \mathcal{F}[\lambda]$, $i = 1, 2, \ldots, r$, $\mathcal{I}_i$ be the principal ideal generated by $a_i(\lambda)$. Then the generating element $d(\lambda)$ of the ideal

$$\mathcal{I} = \mathcal{I}_1 + \mathcal{I}_2 + \cdots + \mathcal{I}_r$$

- the great common devisor

is referred to as the greatest common divisor of $a_1(\lambda)$, $a_2(\lambda)$, ..., $a_r(\lambda)$, denoted

$$d(\lambda) = \text{g.c.d}\{a_1(\lambda), a_2(\lambda), \ldots, a_r(\lambda)\}.$$

If $\text{g.c.d}\{a_1(\lambda), a_2(\lambda), \ldots, a_r(\lambda)\} = 1$, then we say that $a_1(\lambda)$, $a_2(\lambda)$, ..., $a_r(\lambda)$ are coprime.

On the other hand, the generating element $a(\lambda)$ of

$$\mathcal{I} = \mathcal{I}_1 \cap \mathcal{I}_2 \cdots \cap \mathcal{I}_r$$

- the least common multiplier

is said to be the least common multiplier of $a_1(\lambda)$, $a_2(\lambda)$, ..., $a_r(\lambda)$, denoted

$$a(\lambda) = \text{l.c.m}\{a_1(\lambda), a_2(\lambda), \ldots, a_r(\lambda)\}.$$

If $\text{l.c.m}\{a_1(\lambda), a_2(\lambda), \ldots, a_r(\lambda)\} = \prod_{i=1}^{r} a_i(\lambda)$, then we say that $a_1(\lambda)$, $a_2(\lambda)$, ..., $a_r(\lambda)$ are pairwise coprime.

2.3 Ideals of polynomials

- An important theorem

Theorem 2.11 The following statements are equivalent:

1. $a_1(\lambda)$, $a_2(\lambda)$, ..., $a_r(\lambda) \in \mathcal{F}[\lambda]$ are coprime;

2. The ideal generated by $a_1(\lambda)$, $a_2(\lambda)$, ..., $a_r(\lambda)$ is the unit ideal $\mathcal{F}[\lambda]$;

3. there exist $b_1(\lambda)$, $b_2(\lambda)$, ..., $b_r(\lambda) \in \mathcal{F}[\lambda]$ such that

$$a_1(\lambda)b_1(\lambda)+a_2(\lambda)b_2(\lambda)+\cdots+a_r(\lambda)b_r(\lambda) = 1$$

- Example 2.17 on Pg60.

2.4 Factorization of a polynomial

Z. Song

- A polynomial can be decomposed into the product of a new factors-prime
- First, introduce prime and the related properties

Definition 2.8 $a(\lambda) \in \mathcal{F}[\lambda]$ is said to be reducible, if there exist $a_i(\lambda) \in \mathcal{F}[\lambda]$ with $\deg a_i(\lambda) \geq 1$, $i = 1, 2$, such that $a(\lambda) = a_1(\lambda)a_2(\lambda)$. Otherwise $a(\lambda)$ is irreducible. An irreducible polynomial $a_i(\lambda) \in \mathcal{F}[\lambda]$ with $\deg a(\lambda) \geq 1$ is said to be prime in $\mathcal{F}$.

Theorem 2.12 Let $a(\lambda)$, $b(\lambda)$, $c(\lambda) \in \mathcal{F}[\lambda]$ with $a(\lambda)$ being prime and $a(\lambda)|\,[b(\lambda)c(\lambda)]$, then either $a(\lambda)|b(\lambda)$, or $a(\lambda)|c(\lambda)$.

Corollary 2.4 Let $\mathcal{F}$ be a field, $a(\lambda)$, $b_1(\lambda)$, $b_2(\lambda)$, ..., $b_m(\lambda) \in \mathcal{F}[\lambda]$ with $a(\lambda)$ prime in $\mathcal{F}[\lambda]$ and $a(\lambda)|[b_1(\lambda)b_2(\lambda)\cdots b_m(\lambda)]$, then there exists at least one $k \leq m$ such that $a(\lambda)|b_k(\lambda)$.

2.4 Factorization of a polynomial

Z. Song

- Polynomial decomposition

Theorem 2.13 Let $\mathcal{F}$ be a field, $a(\lambda) \in \mathcal{F}[\lambda]$ be monic. Then $a(\lambda)$ can be represented as the product of a few monic prime polynomials. Ignoring the order of the prime polynomials in the product, the prime factorization is unique.

2.4 Factorization of a polynomial

Z. Song

- Polynomial decomposition

Corrollary 2.5 Let $\mathcal{F}$ be a field, $a(\lambda) \in \mathcal{F}[\lambda]$ be monic. Then $a(\lambda)$ can be factorized into the product of the powers of some monic divisors

$$a(\lambda) = [b_1(\lambda)]^{n_1}[b_2(\lambda)]^{n_2} \cdots [b_t(\lambda)]^{n_t}, \quad (1)$$

where $b_i(\lambda) \in \mathcal{F}[\lambda]$ are prime, $n_i \in \mathcal{Z}_+$ for all $i = 1, 2, \ldots, t$, and $b_i(\lambda) \neq b_j(\lambda)$ $\forall i \neq j$. Furthermore, ignoring the order of the divisors $[b_i(\lambda)]^{n_i}$ in (1), the factorization is unique.

Theorem 2.14 Let $\mathcal{F}$ be a field, $a(\lambda) \in \mathcal{F}[\lambda]$ be a monic polynomial with the factorization (1). Define

$$a_i(\lambda) = a(\lambda)/[b_i(\lambda)]^{n_i}, \quad i = 1, 2, \ldots, t,$$

then g.c.d$\{a_1(\lambda), a_2(\lambda), \ldots, a_t(\lambda)\} = 1$.

2.4 Factorization of a polynomial

Z. Song

- Algebraically closed field

Definition 2.9 The field $\mathcal{F}$ is referred to as algebraically closed if any prime polynomial with coefficients in $\mathcal{F}$ is of degree one.

- What is the property, how to determine if a filed is algebraically closed field?

2.4 Factorization of a polynomial

- Algebraically closed field- and important theorem

Theroem 2.15 The following statements are equivalent:

1. $\mathcal{F}$ is algebraically closed;
2. any $a(\lambda) \in \mathcal{F}[\lambda]$ with $\deg a(\lambda) \geq 1$ has at least one prime divisor of degree one; 3. $a(\lambda)$ can be represented as the product of polynomials of degree one. If $\deg a(\lambda) = n$, then $a(\lambda)$ has n roots in $\mathcal{F}$;
4. any monic prime polynomial in $\mathcal{F}[\lambda]$ has the form $\lambda - \alpha$, with $\alpha \in \mathcal{F}$;
5. for any monic polynomial $a(\lambda) \in \mathcal{F}[\lambda]$, there exist $\lambda_1, \lambda_2, \ldots, \lambda_s \in \mathcal{F}$ with $\lambda_i \neq \lambda_j$ for all $i \neq j$ and positive integers $n_1, n_2, \ldots, n_s$ with $n_1 + n_2 + \cdots + n_s = \deg a(\lambda)$ such that

$$a(\lambda) = (\lambda - \lambda_1)^{n_1}(\lambda - \lambda_2)^{n_2} \cdots (\lambda - \lambda_s)^{n_s}. \quad (1)$$

Ignoring the order of the divisors in (1), the factorization is unique.

2.4 Factorization of a polynomial

Z. Song

- Two important theorems

Theroem 2.16 (Basic Theorem in algebra) The field $\mathcal{C}$ of all complex numbers is algebraically closed.

Theroem 2.17 There are two kinds of prime polynomials in $\mathcal{R}[\lambda]$. One is of the form $\lambda - \alpha_i$ with $\alpha_i \in \mathcal{R}$, another one is of the form $\lambda^2 + 2\beta\lambda + \delta$ with $\beta^2 - \delta < 0$. Hence, the roots of $a(\lambda) \in \mathcal{R}[\lambda]$ fall into two categories, one contains all real roots, the other one contains all the complex roots appearing pairwise. Hence, the roots of a polynomial with real coefficients are distributed symmetrically with respect to the real axis.

2.4 Factorization of a polynomial

Z. Song

- Two important theorems

Theroem 2.16 (Basic Theorem in algebra) The field $\mathcal{C}$ of all complex numbers is algebraically closed.

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2.5 Matrix polynomials

- Definition

Definition 2.10(a) Let $a_{ij}(\lambda)$, $i = 1, 2, \ldots, n$, $j = 1, 2, \ldots, m$ be polynomials with coefficients in $\mathcal{F}$ ($\mathcal{C}$ or $\mathcal{R}$), the $m \times n$ matrix $A(\lambda)$ with $a_{ij}(\lambda)$ as its elements in the form

$$A(\lambda) = \begin{bmatrix} a_{11}(\lambda) & a_{12}(\lambda) & \ldots & a_{1n}(\lambda) \\ a_{21}(\lambda) & a_{22}(\lambda) & \ldots & a_{2n}(\lambda) \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}(\lambda) & a_{m2}(\lambda) & \ldots & a_{mn}(\lambda) \end{bmatrix}$$

is called a matrix polynomial or a λ-matrix. The set of all $m \times n$ matrix polynomials is denoted $\mathcal{F}^{m \times n}[\lambda]$.

2.2 Matrix polynomials

Definition 2.10(b) The element $a_{ij}(\lambda)$ of $A(\lambda) \in \mathcal{F}^{m\times n}$ can be written as

$$a_{ij}(\lambda) = a_{ij,0} + a_{ij,1}\lambda + \cdots a_{ij,N}\lambda^N \qquad (1)$$

where $N = \max\{\deg a_{ij}(\lambda)\}$ for $i = 1, 2, \ldots, m$, $j = 1, 2, \ldots, n$. Define

$$A_k = \begin{bmatrix} a_{11,k} & a_{12,k} & \cdots & a_{1n,k} \\ a_{21,k} & a_{22,k} & \cdots & a_{2n,k} \\ \vdots & \ddots & \ddots & \vdots \\ a_{m1,k} & a_{m2,k} & \cdots & a_{mn,k} \end{bmatrix}$$

for $k = 0, 1, 2, \ldots, N$, then $A(\lambda)$ can be represented as

$$A(\lambda) = A_0 + A_1\lambda + \cdots + A_{N-1}\lambda^{N-1} + A_N\lambda^N, \qquad (2)$$

where $A_k \in \mathcal{F}^{m\times n}$. With the above notations, for a polynomial $a_{ij}(\lambda)$ with $\deg a_{ij}(\lambda) = p < N$, then $a_{ij,k} = 0$, $k = p+1, p+2, \ldots, N$. From (2) it is clear that $A_N \neq 0$. N is referred to as the degree of $A(\lambda)$, denoted $\deg A(\lambda) = N$. If $A(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ and $\det A_N \neq 0$, then the matrix polynomial $A(\lambda)$ is regular.

2.5 Matrix polynomials

- Remark

It can be readily verified that for an $A(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ with $\deg A(\lambda) = N$ and $\det A_N \neq 0$, $\deg[\det A(\lambda)] = Nn$.

Addition, scalar multiplication and multiplication of λ-matrices are the same as constant matrices.

2.5 Matrix polynomials

- Properties

Theorem 2.18 $\mathcal{F}^{m\times n}[\lambda]$ is an linear space over the field $\mathcal{F}$, $\mathcal{F}^{n\times n}[\lambda]$ is a noncommutative linear algebra over $\mathcal{F}$ with the unit E_n.

Theorem 2.19 Let $A(\lambda)$, $B(\lambda) \in \mathcal{C}^{n\times n}[\lambda]$ with $B(\lambda)$ regular, then there exist unique $Q_r(\lambda)$, $R_r(\lambda) \in \mathcal{C}^{n\times n}[\lambda]$ such that

$$A(\lambda) = Q_r(\lambda)B(\lambda) + R_r(\lambda) \qquad (1)$$

and either $R_r(\lambda) = 0$ or $\deg R_r(\lambda) < \deg A(\lambda)$. Similarly, there exist unique $Q_l(\lambda)$, $R_l(\lambda) \in \mathcal{C}^{n\times n}[\lambda]$ such that

$$A(\lambda) = B(\lambda)Q_l(\lambda) + R_l(\lambda) \qquad (2)$$

and either $R_l(\lambda) = 0$ or $\deg R_l(\lambda) < \deg A(\lambda)$.

- Proof is on Pg 64-65.

2.6 Unimodular matrix polynomials

Z. Song

- Definition of rank

Definition 2.11 A λ-matrix $A(\lambda)$ is of rank r, written rank $A(\lambda) = r$, if it has a nonzero minor of order r ($r \geq 1$) with all minors of order $r+1$ (if any) being zero. A zero matrix is of rank zero.

- In linear algebra, a **minor** of a matrix ***A*** is the determinant of some smaller square matrix, cut down from ***A*** by removing one or more of its rows or columns. Minors obtained by removing just one row and one column from square matrices (**first minors**) are required for calculating matrix cofactors, which in turn are useful for computing both the determinant and inverse of square matrices.

2.6 Unimodular matrix polynomials

Z. Song

- Definition of unimodular matrix polynomials

Definition 2.12 $A(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ is said to be invertible if there exists a λ-matrix $B(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ such that

$$A(\lambda)B(\lambda) = B(\lambda)A(\lambda) = E_n \qquad (1)$$

$B(\lambda)$ is called the inverse of $A(\lambda)$ and denoted $B(\lambda) = A^{-1}(\lambda)$. In other words, a λ-matrix is invertible if (1) holds for a λ-matrix $B(\lambda)$.

Theorem 2.20 $A(\lambda) \in \mathcal{F}^{n\times n}[\lambda]$ is invertible if and only if $\det A(\lambda)$ is a nonzero scalar in $\mathcal{F}$.

Defintion 2.13 An invertible λ-matrix is called a unimodular matrix.

2.6 Unimodular matrix polynomials

Z. Song

- Definition of elementary operations

Definition 2.14 The following operations on a λ-matrix are called elementary row (column) operations.

1. Interchange the ith row (column) and the jth row (column).

2. Multiply the ith row (column) by a nonzero scalar $\gamma \in \mathcal{F}$.

3. Add to the jth row (column) the ith row (column) multiplied by an arbitrary polynomial $\varphi(\lambda) \in \mathcal{F}[\lambda]$.

2.6 Unimodular matrix polynomials

Z. Song

- Elementary operations

Theorem 2.21 Performing an elementary row operation on an $m \times n$ λ-matrix $A(\lambda)$ is equivalent to premultiplying $A(\lambda)$ by the corresponding elementary matrix in $P(r_i \leftrightarrow r_j)$, $P(\gamma r_i)$ and $P((\varphi)r_i + r_j)$. Similarly, Performing an elementary column operation on an $m \times n$ λ-matrix $A(\lambda)$ is equivalent to postmultiplying $A(\lambda)$ by the corresponding elementary matrix in $P(c_i \leftrightarrow c_j)$, $P(\gamma c_i)$ and $P((\varphi)c_i + c_j)$.

2.6 Unimodular matrix polynomials

Z. Song

- Equivalent matrix polynomials

Definition 2.15 $A(\lambda)$ is said to be equivalent to $B(\lambda)$ if $B(\lambda)$ is obtained by performing a finite series row or/and column operations on $A(\lambda)$, denoted $A(\lambda) \cong B(\lambda)$.

Theorem 2.22 $A(\lambda)$ and $B(\lambda)$ are equivalent if and only if there exist unimodular λ-matrices $P(\lambda)$ and $Q(\lambda)$, such that

$$B(\lambda) = P(\lambda)A(\lambda)Q(\lambda)$$

- Unimodular matrix polynomials are equivalent.

Summary for today

- Ideals of polynomials and properties
- Polynomials factorization

Matrix polynomials

- Invertible matrix polynomials- unimodular-equivalent